

Aliaksei Ivanou | C++ Software Engineer

Qt · Embedded Linux · Yocto · SIP/WebRTC · HMI

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Relocating to Warsaw | 1-month notice | Polish national D visa valid 26 Sep 2026 - 25 Sep 2027

C++ Software Engineer with 6+ years of C and C++, 5 of them in commercial employment, out of 14 years across software engineering and Research & Development. Specialized in Qt-based HMI, Embedded Linux, Yocto/BitBake, NXP i.MX, and SIP/RTP/WebRTC real-time communication. Hands-on experience designing Qt/HMI architecture and delivering multithreaded/asynchronous systems, device integration, networking, Linux debugging, and automated testing, with code reviews, mentoring, onboarding and interview support as technical lead of an engineering sub-group grown from 4 to 6.

Technical Skills

- **C++:** C++11/14/17/20, C (incl. C89), STL, OOP, RAII, smart pointers, object lifetime management, multithreading/concurrency.
- **Architecture & Engineering Practices:** design patterns, MVC/MVVM-style UI architecture, code reviews.
- **Linux:** Embedded Linux, Yocto/BitBake, NXP i.MX, systemd, Linux troubleshooting, service configuration, shell/Bash.
- **Qt/HMI:** Qt 5.15, Qt Widgets, Qt Model/View, Qt signals/slots, QObject, QThread/QtConcurrent, queued connections, QSS.
- **Networking & Real-Time Media:** TCP/IP, SIP, RTP/SDP, WebRTC, WebSockets, liblinphone, mediasoup, SNMP, DHCP, TFTP.
- **Build, Debugging & CI/CD:** CMake, GCC, gdb, Valgrind, core dump analysis, Git, GitLab CI, Jenkins, SSH.
- **Testing:** Google Test, QTest, CTest, unit/integration testing, headless on-target test execution.
- **Databases:** PostgreSQL, SQLite, MySQL.
- **Other:** C#/.NET 8, JavaScript, Java/Scala.

Experience

C++ Software Engineer | [Innowise Group](#)

12/2024 - Present

Engineering services; main assignment to date (05/2025–08/2026): an embedded desk-phone platform for a network-connectivity equipment manufacturer – switches, routers, PBX systems and SIP telephony.

- Completed an internal refactor of the C++17/Qt Widgets SIP call flow into an MVC/MVVM-style architecture, decoupling UI state, user actions, media negotiation, and Linphone SDK integration to simplify feature development and regression testing.
- Implemented active/backup SIP registration and failover/failback logic in Linphone SDK/liblinphone - registration watchdog, realm-aware authentication, AuthInfo synchronization and server-state persistence - so a deployed phone keeps working when the primary SIP server becomes unavailable and returns to it automatically once it recovers.
- Developed and hardened asynchronous SIP/WebRTC calling and video-conferencing flows, including multi-call handling, remote hangups, conference management, media switching, participant management, and audio/video controls.
- Implemented SNMP/DHCP/TFTP/SQLite device-provisioning workflows in a WinForms/.NET 8 utility, including TFTP delivery, DHCP option 66/67 flows, SIP configuration generation, HA1/MD5 authentication, and realm probing.
- Integrated WebRTC/mediasoup media services and the detachable camera/video stack on NXP i.MX Embedded Linux, including Yocto/BitBake customization, systemd configuration, and target-side deployment/debugging.
- Built QTest unit/integration coverage for asynchronous SIP, media and networking workflows, running headless directly on the embedded target so hardware-dependent behavior is verified where it actually runs.
- Technical lead of an internal engineering sub-group that grew from 4 to 6 engineers, alongside ~80% hands-on delivery: code reviews, task planning, onboarding, mentoring and interview support.

Tech Stack: C++17, Qt 5.15.7, Qt Widgets, QtConcurrent/QThread, Embedded Linux, Linphone SDK, SIP/WebRTC, mediasoup, Yocto/BitBake, NXP i.MX, systemd, QTest, C#/.NET 8, SQLite, CMake, GCC, gdb.

Promoted to middle grade after 8 months; relocated from EPAM Belarus to EPAM Poland in 08/2022 without changing project, customer or codebase.

- Developed and maintained C/C++ modules across two production enterprise systems: Digital Library (C89) and Energy ERP (C++17), and performed peer code reviews in distributed international engineering teams.
- Digital Library: Diagnosed production defects on AWS EC2 Linux hosts using SSH, gdb and core-dump analysis; Energy ERP: Diagnosed cross-layer defects in Azure-hosted services from application logs.
- Digital Library: Over 2 years, one of 2-3 C developers maintaining legacy C89 modules in a decades-old platform whose vendor serves 9,000+ libraries worldwide; resolved integration defects between the C core and Java/Scala services under strict backward-compatibility constraints.
- Energy ERP: Implemented C++17 backend features for a Fortune 500 oilfield-services company, tracing data-correctness defects across parsing, MySQL persistence, and JavaScript-facing workflow.
- Maintained JavaScript integration modules connecting backend data flows with the Web UI.

Tech Stack: C89, C++17, PostgreSQL, MySQL, Java 8/11, Scala, JavaScript, Linux, gdb, AWS/Azure, Jenkins.

Independent C++ Software Engineer**05/2020 - 09/2021**

Self-directed transition from MATLAB R&D into commercial C++, alongside EPAM's C++ training program (training, not employment).

- Completed EPAM's C++ mentoring program with an assigned mentor and its C++ laboratory; hired by EPAM at the end of it.
- Built a C++20 desktop application from scratch over 305 commits in four layers - domain models, data access over SQLite, business logic, FLTK UI - with multi-currency accounts over a base currency and exchange-adjusted transfers, a GoogleTest suite wired into CTest, structured logging and a CMake build. Public: github.com/aliaksei-ivanou-by/Home_Accounting.
- Worked systematically through Stroustrup's Programming: Principles and Practice Using C++ (2nd ed.) - fundamentals, OOP, templates and the standard library - building the graphical exercises on FLTK and carrying that drawing layer into the application above. Public: github.com/aliaksei-ivanou-by/Stroustrup_Programming (141 commits).

Tech Stack: C++20, magic_enum, FLTK, SQLite, GoogleTest, CTest, plog, CMake, Git.

C++ Software Engineer | RIFTEK**03/2020 - 05/2020**

Industrial metrology and 3D scanning software.

- Developed a 3D reconstruction R&D prototype for the company's own multi-sensor laser scanners.
- Transformed 2D profile data from several laser scanners into a common coordinate frame and assembled it into a unified 3D point cloud and model.

Tech Stack: C++17, Qt, PCL, Open3D, CMake, Visual Studio, Git.

Design & Research Engineer, Image-Processing Algorithms | PELENG**08/2012 - 01/2020**

Civil Earth-observation and airborne imaging R&D at an optoelectronic instrument manufacturer.

- Developed Earth-observation and airborne image-processing algorithms, including feature-based image stitching, stabilization, debayering, and sensor fusion.
- Implemented custom lossy satellite-image compression, achieving a 3-4x compression ratio.
- Developed image-type prediction and adaptive quantization, improving delivered image quality at the same compression ratio and the same onboard computation budget.
- Validated the imaging chain against real data across 30+ airborne flight tests and satellite in-orbit testing, operating the optical payload and planning test imaging with Systems Tool Kit.

Tech Stack: MATLAB, Systems Tool Kit (STK), PostgreSQL, MS Access, Git.

Education

- Radiophysics, 5-year specialist degree (qualification: Radiophysicist), specialization: Satellite Information Systems and Technologies, [Belarusian State University, Faculty of Radiophysics and Computer Technologies](#) | 2007 - 2012, diploma and transcript apostilled 2025

Languages

English B2 · Polish B1+ · Swedish A2 · Russian Native · Belarusian Native